

Technical Documentation

Book Management System

(React + NestJS)

Project Overview

This project is a Book Management API built using NestJS, Prisma ORM and MySQL. The system supports user authentication using JWT and provides functionality for managing books. Users can register, authenticate and perform CRUD operations on book records associated with their accounts

Technologies Used

Frontend: React (assumed from project description)

Backend: NestJS

ORM: Prisma ORM

Database: MySQL (dev.db)

Authentication: JWT (JSON Web Token)

Password Security: bcrypt hashing

Configuration Management: Environment Variables (.env)

Formatting: Prettier

System Architecture

Presentation Layer (React Client)

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API Layer (NestJS Controllers)

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Business Logic Layer (Services)

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Data Access Layer (Prisma ORM) ↓

MySQL Database

Database Design

Table: User

- ✧ id (Primary Key)
- ✧ email
- ✧ name
- ✧ password- createdAt
- ✧ updatedAt

Table: Book

- ✧ id (Primary Key)
- ✧ title
- ✧ author
- ✧ isbn
- ✧ publishedYear
- ✧ description
- ✧ ownerId (Foreign Key → User.id)
- ✧ createdAt
- ✧ updatedAt

API Endpoints

Authentication

POST /auth/register – Register user POST /auth/login – Login user

Books

GET /books – Retrieve all books

GET /books/:id – Retrieve a specific book

POST /books – Create book

PUT /books/:id – Update book

DELETE /books/:id – Delete book

Authentication Flow

1. User submits registration details.
2. Password is hashed using bcrypt.
3. User record is stored in SQLite via Prisma.
4. User logs in with email and password.
5. Credentials are validated.
6. JWT token is generated.
7. Token is sent to client.
8. Client includes Bearer token in subsequent requests.
9. JWT Guard validates requests before accessing protected resources.

Installation Instructions

- i. Install Node.js and npm.
- ii. Clone the project repository.
- iii. Run: npm install
- iv. Configure environment variables.
- v. Run Prisma migration commands.
- vi. Start development server.

Setup Instructions

Environment Variables:

DATABASE_URL=file:./dev.db

JWT_SECRET=*****

JWT_EXPIRATION=24h PORT=3000

Commands:

npm install

npx prisma migrate dev

npm run start:dev

Challenges Faced

- JWT authentication implementation and token validation.
- Password hashing and security management.
- Database schema design and relationships.
- Prisma migration management.
- API validation and error handling.
- Maintaining clean architecture between modules.

Conclusion

The Book Management API demonstrates a modern full-stack architecture using NestJS, Prisma ORM, MySQL, and JWT authentication. The system follows industry best practices for authentication, database management and API design, making it scalable and maintainable for future enhancements.

